

Symmetries and Conservation HW-2

PROBLEM 1. Derive the non-relativistic Schrödinger equation from the Lagrangian density

$$\mathcal{L} = i\psi^* \frac{\partial \psi}{\partial t} - \frac{1}{2m} |\nabla \psi|^2 - V(\mathbf{x}) |\psi|^2$$

where ψ is a complex field.

1. (10 points) Prove the action is also real, and write down the Hamiltonian density.
2. (10 points) Write $\psi = \frac{1}{\sqrt{2}} (\phi_1 + i\phi_2)$, and $\psi^* = \frac{1}{\sqrt{2}} (\phi_1 - i\phi_2)$, where ϕ_1 and ϕ_2 are two independent real fields. Prove that the two equation of motion for ϕ_1 and ϕ_2 , when combined together, lead to Schrödinger equation.
3. (10 points) By adding "irrelevant" terms that does not change the action, rewrite the Lagrangian density so that it is also real.
4. (10 points) Obviously, this action is $U(1)$ invariant. Using the real Lagrangian density, calculate the associated conserved current. (Hint: the current should be real.)
5. (10 points) Use the real Lagrangian density to derive the energy-momentum tensor $T^\mu{}_\nu$.

SOLUTION:

We first derive the Schrödinger equation using Euler-Lagrange equation:

$$\frac{\partial \mathcal{L}}{\partial \psi} = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \psi)} \right) \implies -V(\vec{x})\psi^* = -\frac{1}{2m} \partial_i (\partial^i \psi^*) + i \frac{\partial \psi^*}{\partial t}$$

Taking complex conjugate for both sides gives:

$$i \frac{\partial \psi}{\partial t} = -\frac{1}{2m} \nabla^2 \psi + V(\vec{x})\psi$$

which is exactly Schrödinger equation.

1.

$S = \int \mathcal{L} d^4x$. $S^* - S = \int d^4x (\mathcal{L}^* - \mathcal{L}) = i \int d^4x (-\psi \frac{\partial \psi^*}{\partial t} - \psi^* \frac{\partial \psi}{\partial t})$. But notice that because we assume that ψ vanish at infinite time boundary, thus by integration by parts,

$$\int d^4x (-\psi \frac{\partial \psi^*}{\partial t} - \psi^* \frac{\partial \psi}{\partial t}) = \int d^3x (-\psi \psi^*|_{t=\infty}) + \int d^4x (\psi^* \frac{\partial \psi}{\partial t}) - \int d^4x (\psi \frac{\partial \psi^*}{\partial t}) = 0$$

which implies $S^* - S = 0$, and thus S is real. □

The Hamiltonian density is given by considering its conjugate momentum:

$$\pi = \frac{\partial \mathcal{L}}{\partial (\dot{\psi})} = i\psi^*$$

$$\mathcal{H} = \pi \dot{\psi} + \pi^* \dot{\psi}^* - \mathcal{L} = -i\psi \frac{\partial \psi^*}{\partial t} + \frac{1}{2} |\nabla \psi|^2 + \frac{1}{2m} V(\vec{x}) |\psi|^2$$

2.

Rewrite the Lagrangian density gives:

$$\begin{aligned}\mathcal{L} &= \frac{i}{2}(\phi_1 - i\phi_2)\frac{\partial(\phi_1 + i\phi_2)}{\partial t} - \frac{1}{4m}[\nabla(\phi_1 + i\phi_2)] \cdot [\nabla(\phi_1 - i\phi_2)] - \frac{1}{2}V(\vec{x})(\phi_1 + i\phi_2)(\phi_1 - i\phi_2) \\ &= \frac{1}{2}(-\phi_1\frac{\partial\phi_2}{\partial t} + \phi_2\frac{\partial\phi_1}{\partial t} + i(\phi_1\frac{\partial\phi_1}{\partial t} + \phi_2\frac{\partial\phi_2}{\partial t})) - \frac{1}{4m}((\nabla\phi_1)^2 + (\nabla\phi_2)^2) - \frac{1}{2}V(\vec{x})(\phi_1^2 + \phi_2^2)\end{aligned}$$

Plug the Lagrangian into the Euler-Lagrange equation:

$$\begin{aligned}-V(\vec{x})\phi_1 + \frac{i}{2}\frac{\partial\phi_1}{\partial t} - \frac{1}{2}\frac{\partial\phi_2}{\partial t} &= \frac{i}{2}\frac{\partial\phi_1}{\partial t} - \frac{1}{2m}\nabla^2\phi_1 + \frac{1}{2}\frac{\partial\phi_2}{\partial t} \implies -\frac{\partial\phi_2}{\partial t} = H\phi_1 \\ -V(\vec{x})\phi_2 + \frac{i}{2}\frac{\partial\phi_2}{\partial t} + \frac{1}{2}\frac{\partial\phi_1}{\partial t} &= \frac{i}{2}\frac{\partial\phi_2}{\partial t} - \frac{1}{2m}\nabla^2\phi_2 - \frac{1}{2}\frac{\partial\phi_1}{\partial t} \implies \frac{\partial\phi_1}{\partial t} = H\phi_2\end{aligned}$$

Adding the above equations gives:

$$i\frac{\partial(\phi_1 + i\phi_2)}{\partial t} = H(\phi_1 + i\phi_2)$$

which is exactly the Schrödinger equation.

3.

Let $\mathcal{L}' = \mathcal{L} - \frac{i}{2}\frac{\partial(\psi\psi^*)}{\partial t}$. Because $\mathcal{L}$ is equivalent under transformation $\mathcal{L} \rightarrow \mathcal{L} + \partial_\mu\alpha^\mu$, thus $\mathcal{L}'$ is equivalent to $\mathcal{L}$. And it is observed that $\mathcal{L}' = \frac{\mathcal{L} + \mathcal{L}^*}{2}$, thus $\mathcal{L}'$ is real.

4.

We have proved in class that the conserved current with respect to $U(1)$ symmetry is

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}i\psi - \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi^*)}i\psi^*$$

This is because when considering the local transformation $\psi \rightarrow (1 + i\theta)\psi$,

$$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\psi}i\theta\psi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\partial_\mu(i\theta\psi) - \frac{\partial\mathcal{L}}{\partial\psi^*}i\theta\psi^* + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi^*)}\partial_\mu(i\theta\psi^*) = \theta\partial_\mu(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}i\psi - \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi^*)}i\psi^*) = 0$$

Then by plugging in $\mathcal{L}'$ we have:

$$j^\mu = (-\psi\psi^*, \frac{1}{2m}(i\psi\frac{\partial\psi^*}{\partial x} - i\psi^*\frac{\partial\psi}{\partial x}), \frac{1}{2m}(i\psi\frac{\partial\psi^*}{\partial y} - i\psi^*\frac{\partial\psi}{\partial y}), \frac{1}{2m}(i\psi\frac{\partial\psi^*}{\partial z} - i\psi^*\frac{\partial\psi}{\partial z}))$$

which gives the conservation current, and it's apparently real.

5.

Consider a spacetime translation: $\delta\mathcal{L} = a^\mu\partial_\mu\mathcal{L}$, $\delta\psi = a^\mu\partial_\mu\psi$, $\delta\psi^* = a^\mu\partial_\mu\psi^*$. Then

$$\delta\mathcal{L} = \partial_\mu(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\delta\psi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi^*)}\delta\psi^*) = \partial_\mu(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}a^\nu\partial_\nu\psi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi^*)}a^\nu\partial_\nu\psi^*) = a^\nu\partial_\nu\mathcal{L}$$

And the above gives the energy momentum tensor of complex scalar field ψ :

$$T^\mu{}_\nu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\partial_\nu\psi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi^*)}\partial_\nu\psi^* - \delta^\mu{}_\nu\mathcal{L}$$

Plugging in $\mathcal{L}'$ gives:

$$\begin{aligned} T^0_0 &= \frac{1}{2m} |\nabla \psi|^2 + V|\psi|^2 \\ T^1_1 &= \frac{i}{2} \psi \frac{\partial \psi^*}{\partial t} - \frac{i}{2} \psi^* \frac{\partial \psi}{\partial t} + V|\psi|^2 + \frac{1}{2m} (-|\frac{\partial \psi}{\partial x}|^2 + |\frac{\partial \psi}{\partial y}|^2 + |\frac{\partial \psi}{\partial z}|^2) \\ T^2_2 &= \frac{i}{2} \psi \frac{\partial \psi^*}{\partial t} - \frac{i}{2} \psi^* \frac{\partial \psi}{\partial t} + V|\psi|^2 + \frac{1}{2m} (|\frac{\partial \psi}{\partial x}|^2 - |\frac{\partial \psi}{\partial y}|^2 + |\frac{\partial \psi}{\partial z}|^2) \\ T^3_3 &= \frac{i}{2} \psi \frac{\partial \psi^*}{\partial t} - \frac{i}{2} \psi^* \frac{\partial \psi}{\partial t} + V|\psi|^2 + \frac{1}{2m} (|\frac{\partial \psi}{\partial x}|^2 + |\frac{\partial \psi}{\partial y}|^2 - |\frac{\partial \psi}{\partial z}|^2) \\ T^0_i &= \frac{i}{2} (\psi^* \partial_i \psi - \psi \partial_i \psi^*) \\ T^i_0 &= \frac{1}{2m} (\partial_i \psi^* \partial_t \psi + \partial_i \psi \partial_t \psi^*) \end{aligned}$$

PROBLEM 2. Unit System: In the units of $c = \hbar = 1$,

1. Find the length (in meters) and time (in seconds) corresponding to 1/ gram. (10 points)
2. Also, in this unit system, find the relation between 1 gram and 1 eV . (10 points)

SOLUTION: 1.

$$\hbar = 1 \implies 1.0546 \times 10^{-34} J = 1 s^{-1}, c = 1 \implies 3 \times 10^8 m = 1 s$$

Thus

$$\begin{aligned} g^{-1} &= 1000 kg^{-1} = 1000 (J \cdot s^2 / m^2)^{-1} = 1000 \cdot \frac{s^2}{9 \times 10^{16}} \cdot 1.0546 \times 10^{-34} s \cdot s^{-2} \\ &= 1.17 \times 10^{-48} s = 3.52 \times 10^{-40} m \end{aligned}$$

2.

$$\begin{aligned} 1 eV &= 1.602 \times 10^{-19} J = 1.519 \times 10^{15} s^{-1} \\ 1 g &= 8.547 \times 10^{47} s^{-1} \implies 1 g = 5.627 \times 10^{32} eV \end{aligned}$$

PROBLEM 3. Special relativity: Introduce the antisymmetric rank two tensor

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

where $A^\mu = (\varphi, \mathbf{A})$ is a four-dimensional vector potential. Here, we choose the metric as $(1, -1, -1, -1)$.

1. (10 points) Express F^{0i} and F^{ij} in terms of the electrical and magnetic fields, $\mathbf{E}$ and $\mathbf{B}$.
2. (20 points) Is the following formula for Lorentz force Lorentz invariant?

$$\begin{aligned} \frac{d\mathbf{p}}{dt} &= q \left(\mathbf{E} + \frac{1}{c} \mathbf{v} \times \mathbf{B} \right), \quad \frac{dE}{dt} = q \mathbf{E} \cdot \mathbf{v} \\ \text{with } \mathbf{v} &= \frac{d\mathbf{x}}{dt}, \quad \mathbf{p} = \gamma \mathbf{v}, \quad \gamma = (1 - v^2)^{-\frac{1}{2}} \end{aligned}$$

Show your calculation instead of simply arguing by words. No matter whether Eq. 1 is Lorentz invariant or not, write down your version of Lorentz force, so that is obviously Lorentz invariant.

SOLUTION:

1.

I suppose here $i, j = 1, 2, 3$.

$$\begin{aligned} F^{0i} &= \partial_t A^i + \frac{\partial \varphi}{\partial x^i} = -E^i \\ F^{ij} &= \partial^i A^j - \partial^j A^i = \epsilon_{ijl} \partial^i A^j \vec{e}^l \cdot \vec{e}^k = \epsilon_{ijk} (\nabla \times \vec{A})_k = \epsilon_{ijk} B^k, \text{ where } \epsilon \text{ is the Levi-Civita symbol.} \end{aligned}$$

2.

It suffices to consider a standard boost of velocity u over x axis. Then the Lorentz transformation matrix is:

$$\Lambda^\mu{}_\nu = \begin{bmatrix} \gamma & -\gamma u & 0 & 0 \\ -\gamma u & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Because $F^{\mu\nu}$ is a Lorentz tensor,

$$\begin{aligned} F^{\mu'\nu'} &= \Lambda^{\mu'}{}_\mu \Lambda^{\nu'}{}_\nu F^{\mu\nu} \\ \implies E'_x &= E_x, E'_y = \gamma(E_y - uB_z), E'_z = \gamma(E_z + uB_y) \\ \implies B'_x &= B_x, B'_y = \gamma(B_y + uE_z), B'_z = \gamma(B_z - uE_y) \end{aligned}$$

Because p^μ is also a Lorentz tensor,

$$p^{\mu'} = \Lambda^{\mu'}{}_\mu p^\mu \implies p'_x = \gamma(p_x - uE), p'_y = p_y, p'_z = p_z, E' = \gamma(E - up_x)$$

Then let's consider LHS and RHS in S' frame, firstly let's consider x axis:

$$\begin{aligned} \frac{dp'_x}{dt'} &= \frac{d(\gamma(p_x - uE))}{dt} \frac{1}{\frac{dt'}{dt}} = \frac{(\frac{dp_x}{dt} - u\frac{dE}{dt})}{1 - uv_x} = \frac{qE_x + qv_yB_z - qv_zB_y - qu(v_xE_x + v_yE_y + v_zE_z)}{1 - uv_x} \\ q(E'_x + (v'_yB'_z - v'_zB'_y)) &= qE_x + q\frac{v_y}{1 - uv_x}(B_z - uE_y) - q\frac{v_z}{1 - uv_x}(B_y + uE_z) \end{aligned}$$

It then can be observed that $\frac{dp'_x}{dt'} = q(E'_x + (v'_yB'_z - v'_zB'_y))$.

Next let's prove the y axis:

$$\begin{aligned} \frac{dp'_y}{dt'} &= \frac{\frac{dp_y}{dt}}{\gamma(1 - uv_x)} = \frac{qE_y - qv_xB_z + qv_zB_x}{\gamma(1 - uv_x)} \\ q(E'_y - (v'_xB'_z - v'_zB'_x)) &= q\gamma(E_y - uB_z) - \frac{q\gamma(v_x - u)(B_z - uE_y)}{1 - uv_x} + \frac{qv_zB_x}{\gamma(1 - uv_x)} \end{aligned}$$

Then to see they are equal, it suffices to observe that:

$$qE_y - qv_xB_z = q\gamma^2(E_y - uB_z)(1 - uv_x) - q\gamma^2(v_x - u)(B_z - uE_y)$$

This is because RHS equals

$$\frac{E_y - uB_z - uv_xE_y + u^2v_xB_z - v_xB_z + uv_xE_y + uB_z - u^2E_y}{1 - u^2} = E_y - v_xB_z$$

We can prove the z axis in the same fashion as y . (y, z are essentially symmetric here)

Finally let's prove the energy term:

$$\begin{aligned} \frac{dE'}{dt'} &= \frac{\frac{dE}{dt} - u\frac{dp_x}{dt}}{1 - uv_x} = \frac{q(v_xE_x + v_yE_y + v_zE_z) - u(qE_x + qv_yB_z - qv_zB_y)}{1 - uv_x} \\ q(v'_xE'_x + v'_yE'_y + v'_zE'_z) &= q\frac{(v_x - u)E_x + v_y(E_y - uB_z) + v_z(E_z + uB_y)}{1 - uv_x} \end{aligned}$$

It is apparent that they are the same.

Finally we have proved that the formula is Lorentz invariant. □

It can be observed directly by turning the formula into:

$$\frac{dp^\mu}{d\tau} = qF^{\mu\nu}u_\nu$$

where u_ν is the 4-velocity $u_\nu = (\gamma, -\gamma\vec{v})$, τ is the proper time.